

Mango: the block override language

Quick reference. See the *README* for the rest of the plugin.

Every lane has one set of knobs, but **each block can override them with a line of text**. That is how one delay lane plays a different time in every block, or a filter tracks the note you just played. Select a block and type into the **block overrides** field at the bottom right.

Syntax

key=value key=value key=value

Space-separated **key=value** pairs, up to 16 per block. Newlines count as spaces (press **Ctrl+Return** for a newline, **Return** to commit).

- Each effect reads only the keys it knows and ignores the rest. The panel above the field lists the keys the selected lane understands.
- A key you set here overrides that knob **for this block only**.
- **All or nothing:** if any part of the line does not parse, *none* of it is applied and the field turns red. Your text is kept so you can fix it.

Values

Form	Meaning	Example
number	a plain value	fb=0.6
mididur	length of one cycle of the last MIDI note, in seconds	dur=mididur
midifreq	pitch of that note, in Hz (= 1/mididur)	f0=midifreq
voice digit	the same, on one note of the chord you are holding	dur=mididur2, f0=midifreq1
s4 s8 s16 s32 s64	straight note-value shortcut (fraction of a whole note)	dur=s8
t4 t8 t16 t32 t64	triplet note-value shortcut	dur=t8
d4 d8 d16 d32 d64	dotted note-value shortcut	dur=d8
number + s	an absolute time in seconds , ignores tempo and mididur	dur=1.5s
scaled	a magic word or note-value shortcut, times or divided by a number	mididur*2, s4/3, 2*d16
vowel letter	for v0 / v1 only	v0=a

MIDI tracking. Without a digit both follow the last note-on, sampled when the block starts: **dur=mididur** tunes a delay to the note, **f0=midifreq** points a filter at it.

Chords. Add a digit **1 to 4** to address a note of the chord currently held, counted **from the lowest up**: **mididur1** is the bottom note, **mididur2** the next one, and so on (**midifreq1..4** likewise). Give each lane a different voice and one chord tunes the whole rack:

```
lane 1   dur=mididur1 fb=0.97      \  
lane 2   dur=mididur2 fb=0.97      )  a chord of tuned combs  
lane 3   dur=mididur3 fb=0.97      /
```

Up to four notes are followed at once; play a fifth and the **oldest** is dropped, so the chord always tracks what you played last. Ask for a voice you are not holding (**mididur3** on a two-note chord) and you get the top note instead of silence, and with nothing held at all you get the last note played, exactly like plain **mididur**.

Note-value shortcuts. **s4 s8 s16 s32 s64** (straight), **t4 t8 t16 t32 t64** (triplet) and **d4 d8 d16 d32 d64** (dotted) are the same fractions the duration-weight grid draws from - **s8** is 0.125, **t8** is $0.125 \times 2/3$, **d8** is 0.125×1.5 - so **dur=t8** beats typing the decimal out. They scale like the magic words do: **s4*2**, **t8/3**, **2*d16**.

Absolute seconds. A number written with a trailing **s** is a time in seconds outright: **dur=1.5s** is 1.5 seconds, independent of both tempo and mididur - the same “already resolved, skip the tempo maths” behaviour **mididur/midifreq** get, just with a fixed value instead of one that follows what you play.

The one unit trap

`dur` changes meaning depending on how you write it:

Written	Means	At 120 bpm
<code>dur=0.25</code>	a fraction of a whole note , follows host tempo	a quarter = 0.5 s
<code>dur=0.125</code>	eighth note	0.25 s
<code>dur=s8</code>	the same eighth, spelled from the weight grid's names	0.25 s
<code>dur=mididur</code>	a time in seconds , ignores tempo	1 cycle of the note
<code>dur=1.5s</code>	a time in seconds , ignores tempo and mididur	1.5 s

Every other key is in the same unit as its knob.

Keys

Key	Controls	Range
<code>dur</code>	rate / time (see above)	whole-note fraction, or seconds
<code>mix</code>	wet/dry of the effect	0–1 (0 = transparent)
<code>att rel</code>	attack / release length	0–1 = fraction of the shaped duration
<code>attcurve relcurve</code>	edge shape	0 slow · 0.5 straight · 1 very fast
<code>fb</code>	delay feedback	0–0.999
<code>damp</code>	delay damping	0–1
<code>porta</code>	delay time glide	1–50 (ms)
<code>model</code>	distortion model	0 Standard · 1 Dynamic · 2 Triode · 3 Class AB
<code>drive</code>	distortion drive	0–40 (dB)
<code>bias</code>	distortion bias	0–0.5
<code>sag</code>	supply sag	0–1
<code>gain</code>	distortion output gain	–24 to +24 (dB)
<code>mode</code>	filter mode	0 Lowpass · 1 Highpass · 2 Formant
<code>mode</code>	panner mode	0 Cycle -> · 1 Cycle <- · 2 Cycle <-> · 3 Random
<code>q</code>	filter resonance	0.3–12
<code>f0 f1</code>	start / end frequency	filter 20–20000, ring 0.5–10000 (Hz)
<code>v0 v1</code>	start / end vowel	a e i o u
<code>bits</code>	bit depth	1–24
<code>down</code>	downsample factor	1–64
<code>amp</code>	ring modulation amount	0–1 (this is the ring's mix)
<code>fade</code>	reverser seam fade	0–0.5
<code>width</code>	freeze stereo width	0–1 (1 wide, 0 mono)
<code>aux1 aux2</code>	aux send levels	0–1
<code>pass</code>	aux main passthrough	0–1 (this is the aux send's mix)
<code>glide</code>	panner travel time	0–1 = fraction of a step
<code>w4 w8 w16 w32 w64</code>	duration weights	0–1 each
<code>wstr wtrip wdot</code>	straight / triplet / dotted weights	0–1 each

`mode` means different things on the filter and the panner (each lane only sees its own).

Which effect reads what

Effect	Keys
Gater	dur att rel attcurve relcurve mix + weights
Grain	dur att rel attcurve relcurve mix + weights
Delay	dur fb damp porta mix
Dist	model drive bias sag gain mix
Filter	dur mode q f0 f1 v0 v1 mix + weights
Quant	bits down mix
Ring	dur f0 f1 amp + weights
Rev	dur fade mix + weights
Freeze	mix width + weights
Aux	dur att rel attcurve relcurve aux1 aux2 pass + weights
Pan	dur mode glide mix + weights

weights = w4 w8 w16 w32 w64 wstr wtrip wdot

Duration weights

Effects with a rhythmic rate do not take a fixed value: you weight how likely each note length is, and the rate is **drawn** at the start of each block. Setting *dur* pins it instead and skips the draw.

w4=1 w8=0 w16=0 w32=0	always a quarter
w16=1 w32=0.5	sixteenths, sometimes thirty-seconds
w32=1 w64=1	a fast, uneven stutter
w8=1 wstr=1 wtrip=1	eighths, straight or triplet

Weights are relative, not percentages: *w4=1 w8=0.5* makes a quarter twice as likely as an eighth. The draw depends only on the seed, the lane and the block, so a pattern always plays back the same way and the block picture shows exactly what you will hear.

Recipes

dur=0.125 fb=0.6	delay: eighth-note repeats, more feedback
dur=mididur fb=0.99 damp=0.4	delay: plucked string tuned by MIDI
f0=midifreq f1=midifreq*2	filter: sweep the note to its octave
v0=a v1=u	filter: glide from "ah" to "oo"
mode=2 v0=i v1=o	filter: force formant mode, I to O
w16=1 w32=1 mix=0.5	gater: fast, half depth
dur=mididur/2 amp=1	ring: carrier at twice the note pitch
f0=midifreq*2 q=8	filter: ring the second note of the chord
bits=3 down=8 mix=0.7	quant: crunchy, blended back
pass=0 aux1=1	aux: send this block away from the main mix
mode=3 glide=0	panner: random, hard jumps
w4=1 wdot=1	freeze: re-capture every dotted quarter
dur=t8 fb=0.5	delay: eighth-note triplet repeats
dur=1.5s mix=1	gater: pinned to 1.5 s, whatever the tempo

Gotchas

- **Red field = nothing applied.** One bad key disables the whole line, not just that pair.
- **Unknown keys are an error**, not ignored: a typo like *feedback=0.6* turns the line red. Keys an effect does not *use* are fine; keys that do not *exist* are not.
- *mix=0* makes the lane transparent, so the block does nothing. On the ring modulator that key is *amp*, on the aux send it is *pass*.
- *dur* has no effect on Freeze: its retrigger rate comes from the weights only.
- A *mididur* expression is in seconds and **ignores the host tempo**. That is the point, but it means *dur=mididur* will not stay in time with the grid unless the note happens to line up.
- Voice digits stop at 4: *mididur5* is a typo, not a fifth voice, so it turns the line red.
- The voice digits read the notes held **when the block starts**. Releasing a note does not retune a block that is already sounding; the next block picks up the new chord.
- Note-value shortcuts and the seconds suffix are exact matches only: *s5* (not a real note value), *dur=s* (no digits) and *dur=1.5sx* are all typos, not something else, and turn the line red like any other bad key.